Lecture #12 | Machine learning

SE377 Introduction to Big Data Analysis and Visualization (2017)

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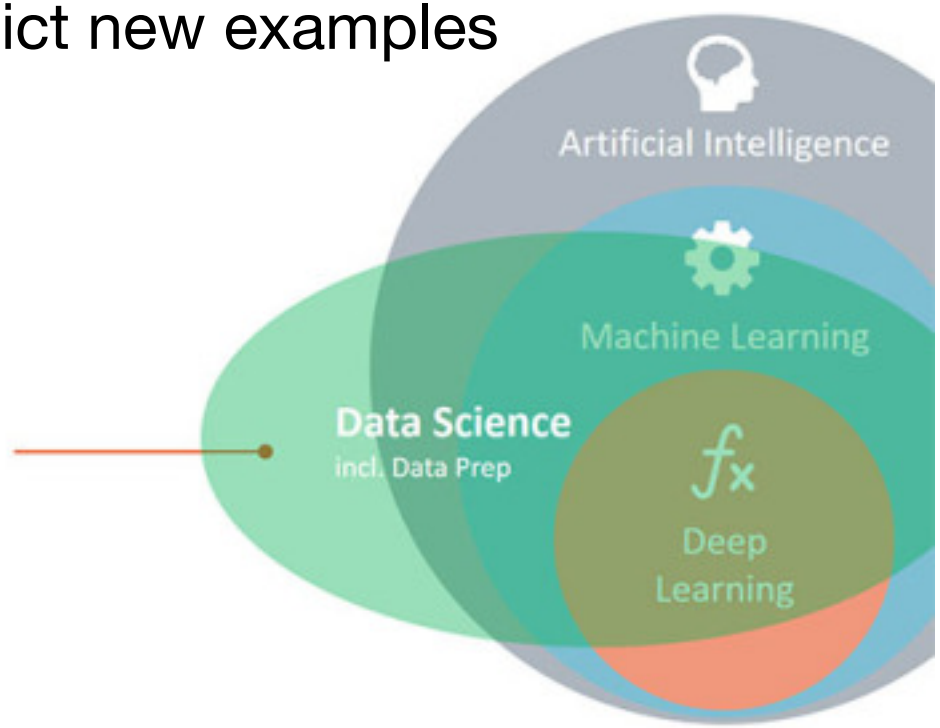
What is machine learning?

- A field of computer science that gives computers the ability to learn without being explicitly programmed
- Machine learning (ML) algorithms can generalize from existing examples of a task so that it can recognize or predict new examples

Data Science

Covers the practical application of advanced analytics, statistics, machine learning, and the necessary data preparation in a business context.



Source: wikipedia, rapidminer

Types of machine learning

- Supervised Learning
  - Training data is labeled
  - Goal is correctly label new data
  - Examples: classification (for categorical data), regression (for continuous data)
  - c.f., semi-supervised learning
- Reinforcement Learning
  - Training data is unlabeled
  - System receives feedback for its actions
  - Goal is to perform better actions
- Unsupervised Learning
  - Training data is unlabeled
  - Goal is to categorize the observations
  - Examples: clustering, outlier detection

Applications of machine learning

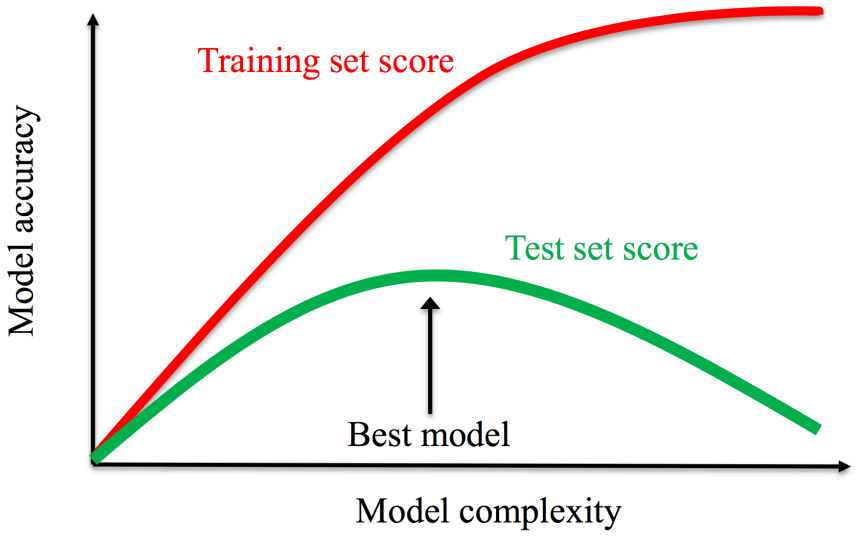
- Handwriting Recognition
  – convert written letters into digital letters
- Language Translation
  – translate spoken and or written languages (e.g. Google Translate)
- Speech Recognition
  – convert voice snippets to text (e.g. Siri, Cortana, and Alexa)
- Image Classification
  – label images with appropriate categories (e.g. Google Photos)
- Autonomous Driving
  – enable cars to drive

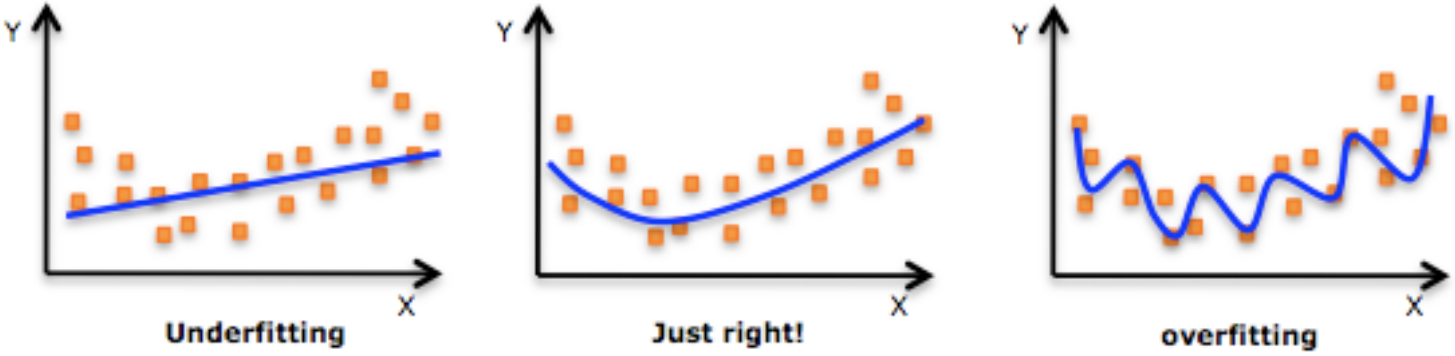
Features in machine learning

- Features are the observations that are used to form predictions
  - For image classification, the pixels are the features
  - For voice recognition, the pitch and volume of the sound samples are the features
  - For autonomous cars, data from the cameras, range sensors, and GPS are features
- Extracting relevant features is important for building a model
  - Time of day is an irrelevant feature when classifying images
  - Time of day is relevant when classifying emails because SPAM often occurs at night
- Common Types of Features in Robotics
  - Pixels (RGB data)
  - Depth data (sonar, laser rangefinders)
  - Movement (encoder values)
  - Orientation or Acceleration (Gyroscope, Accelerometer, Compass)

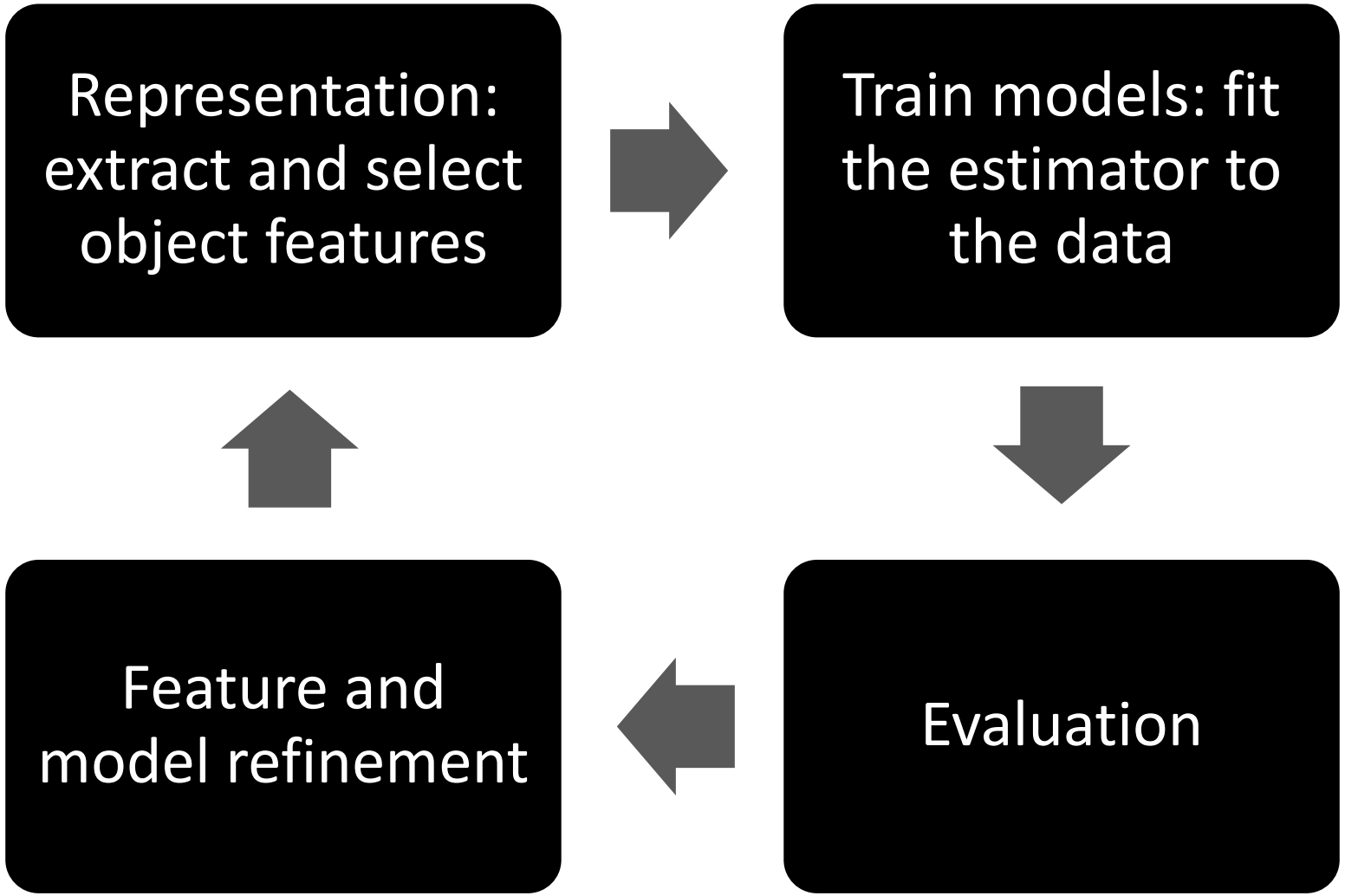
Training and test data

- Training Data: data used to learn a model
- Test Data: data used to assess the accuracy of model
- Overfitting: model performs well on training data but poorly on test data





Represent/Train/Evaluate/Refine cycle



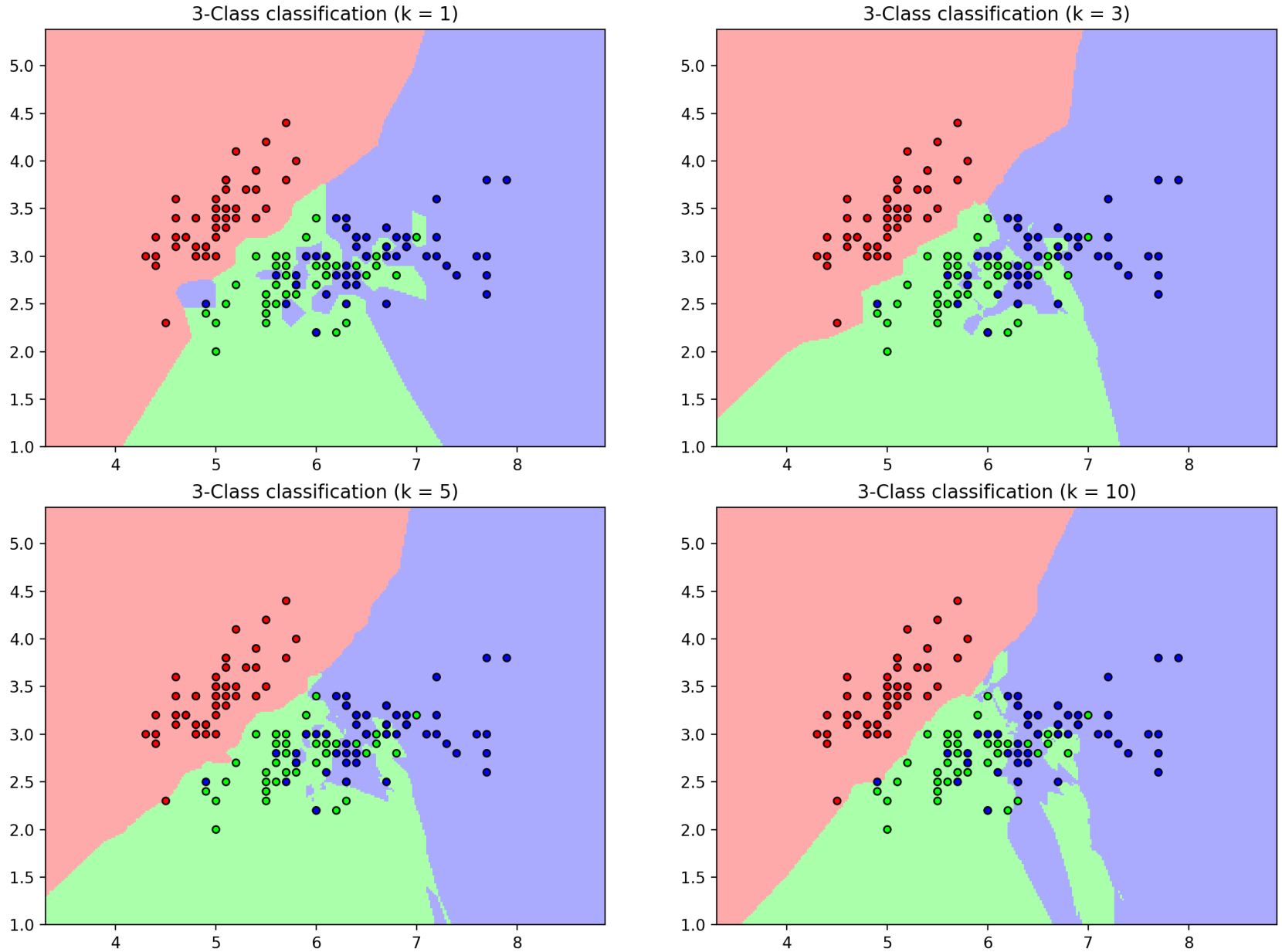
The k-nearest neighbor (k-NN) classifier

- Given a training set X with label y, and given a new instance x to be classified
  1. Find the most similar instances (let's call them X_NN) to x that are in X
  2. Get the labels y_NN for the instances in X_NN
  3. Predict the label for x by combining the labels y_NN (e.g., simple majority vote)

Supervised Learning Algorithms

- K-Nearest Neighbor
- Linear Regression
- Decision Trees
- Support Vector Machines
- Neural Networks

Visual representation of k-NN



k-NN issues

- The Data is the Model
  - No training needed
  - Accuracy generally improves with more data
  - Matching is simple and fast (and single pass)
  - Usually need data in memory, but can be run off disk
- Minimal Configuration
  - Number of neighbors: k
  - Distance metric
  - Weighting function (e.g., inverse distance)
  - Method for aggregating the classes of neighbor points (typically majority vote)

k-NN distance metrics

- Euclidean Distance: simplest, fast to compute (Minkowski with p=2)

$$d(x, y) = \|x - y\|$$

- Cosine Distance: good for documents, images, etc.

$$d(x, y) = 1 - \frac{x \cdot y}{\|x\| \|y\|}$$

- Jaccard Distance: set data

$$d(X, Y) = 1 - \frac{|X \cap Y|}{|X \cup Y|}$$

- Hamming Distance: string data

$$d(x, y) = \sum_{i=1}^{n} (x_i \neq y_i)$$

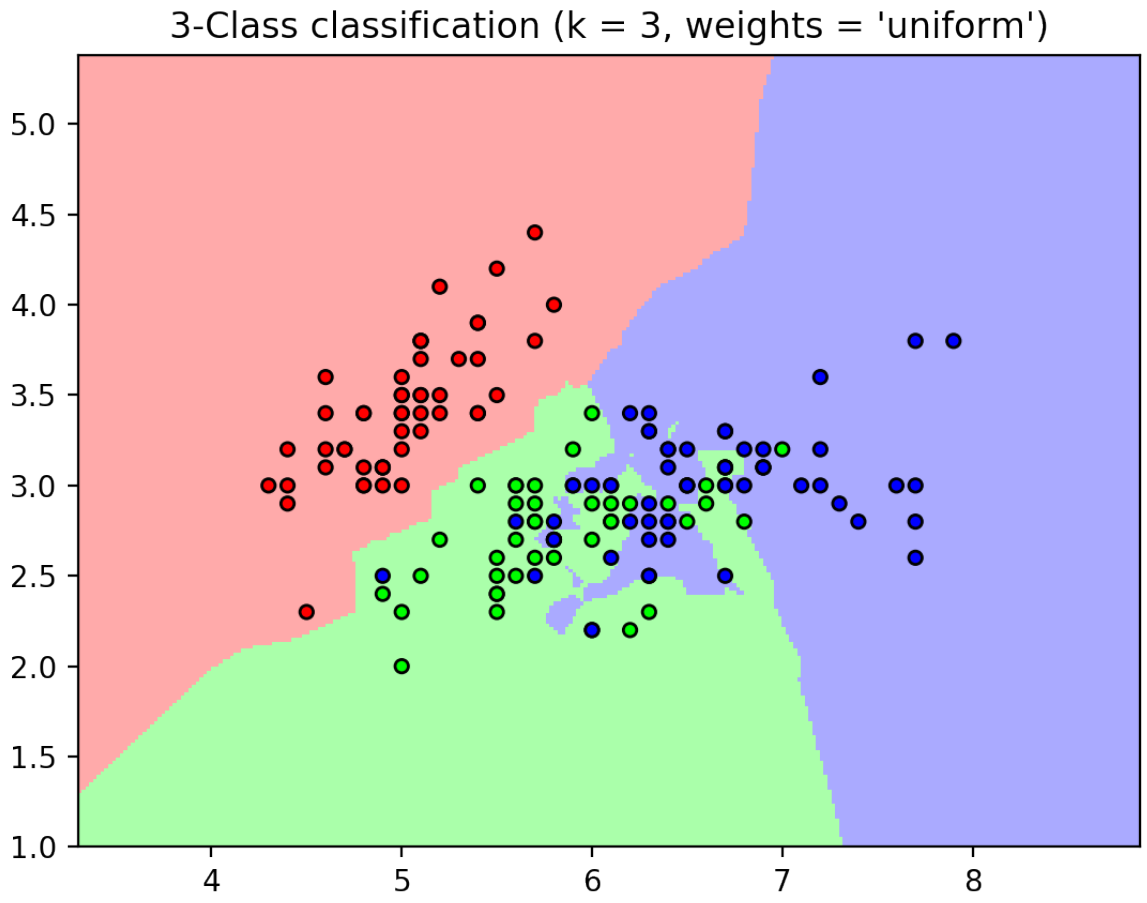
k-NN distance metric (cont.)

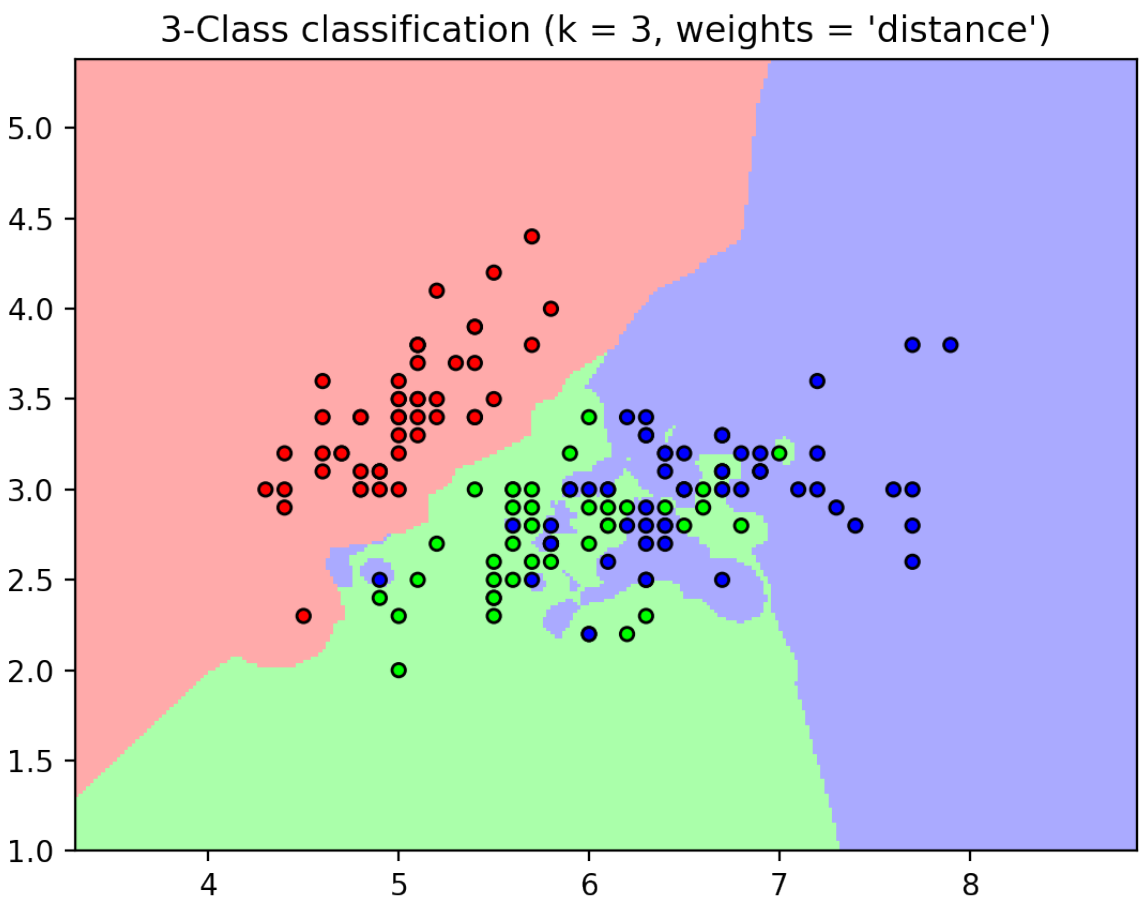
- Manhattan Distance: Coordinate-wise distance

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

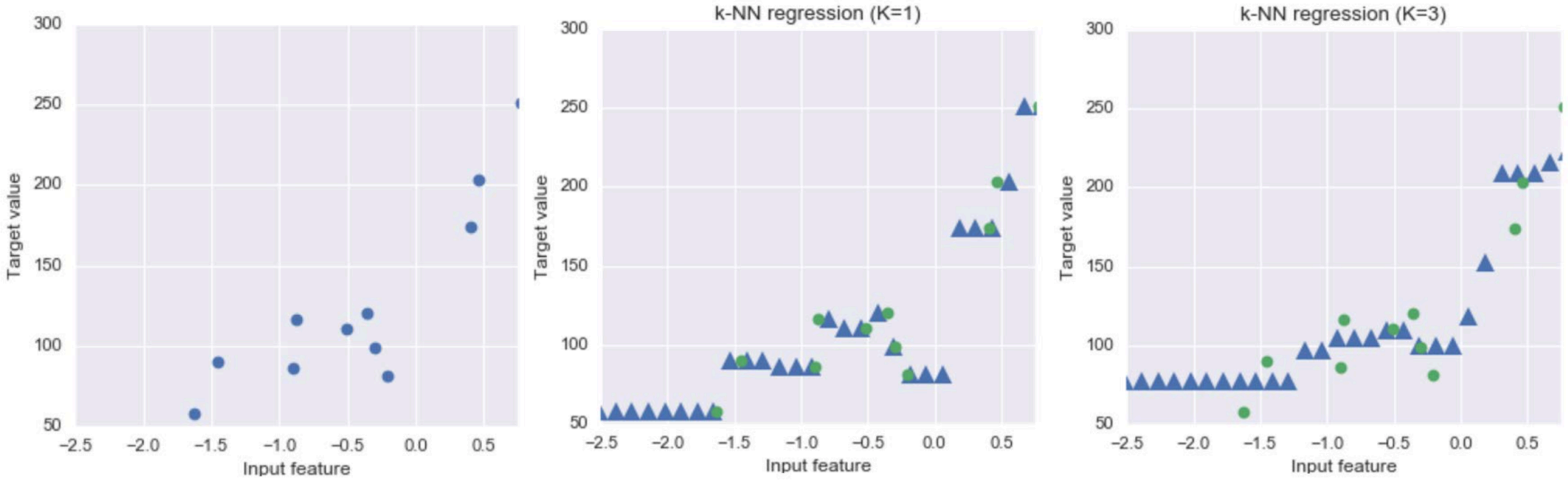
- Edit Distance: for strings, especially genetic data
- Mahalanobis Distance: Normalized by the sample covariance matrix – unaffected by coordinate transformations.

k-NN with different distance metric



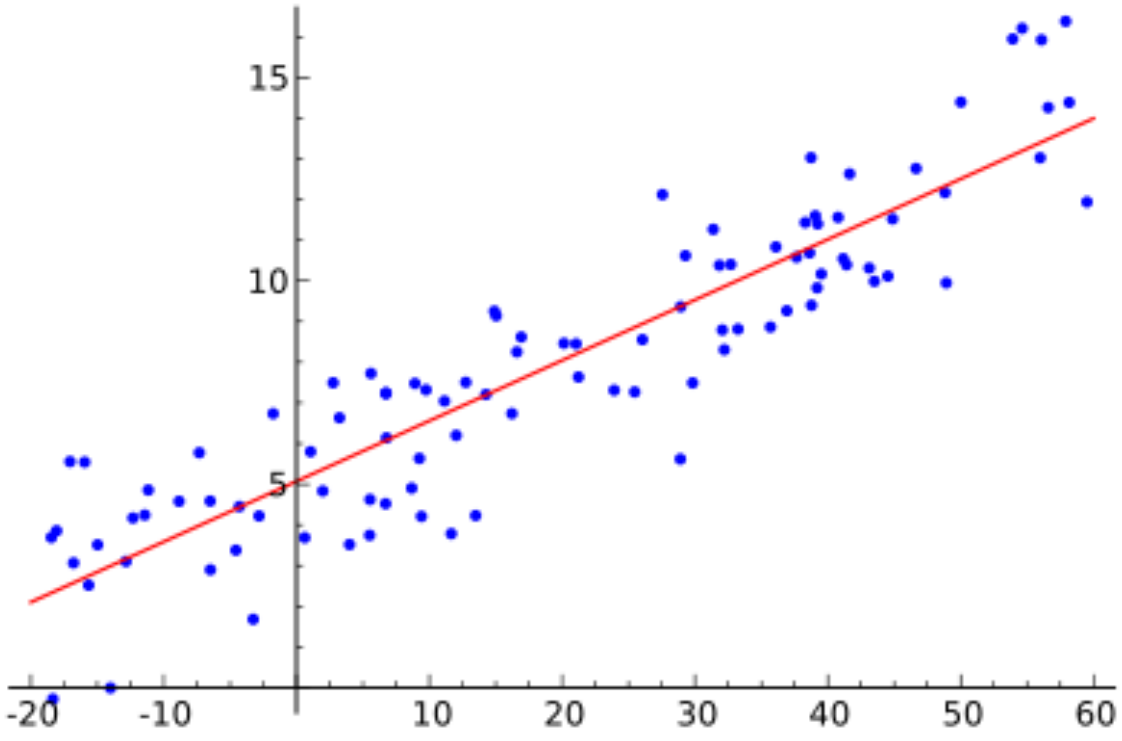


k-Nearest Neighbors Regressions



Linear regressions

- Find the best line (linear function y=f(x)) to explain the given data



Linear regressions (cont.)

- The predicted value of y is given by:

$$\hat{y} = \hat{\beta}_0 + \sum_{j=1}^{p} X_j \hat{\beta}_j$$

- The vector of coefficients $\hat{\beta}$ is the regression model
- If $X_0 = 1$, the formula becomes a matrix product

$$\hat{y} = \mathrm{X}\,\hat{\beta}$$

Residual Sum-of-Squares (RSS)

- To determine the model parameters $\hat{\beta}$ from some data, we can write down the Residual Sum of Squares:

$$\mathrm{RSS}(\beta) = \sum_{i=1}^{N} (y_i - \beta x_i)^2$$

- Or symbolically $\mathrm{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)$
- To minimize it, take the derivative w.r.t. β which gives

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) = 0$$

- And if $\mathbf{X}^T\mathbf{X}$ is non-singular, the unique solution is

$$\hat{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

The R^2 (r-squared) regression score

- Measures how well a prediction model for regression fits the given data

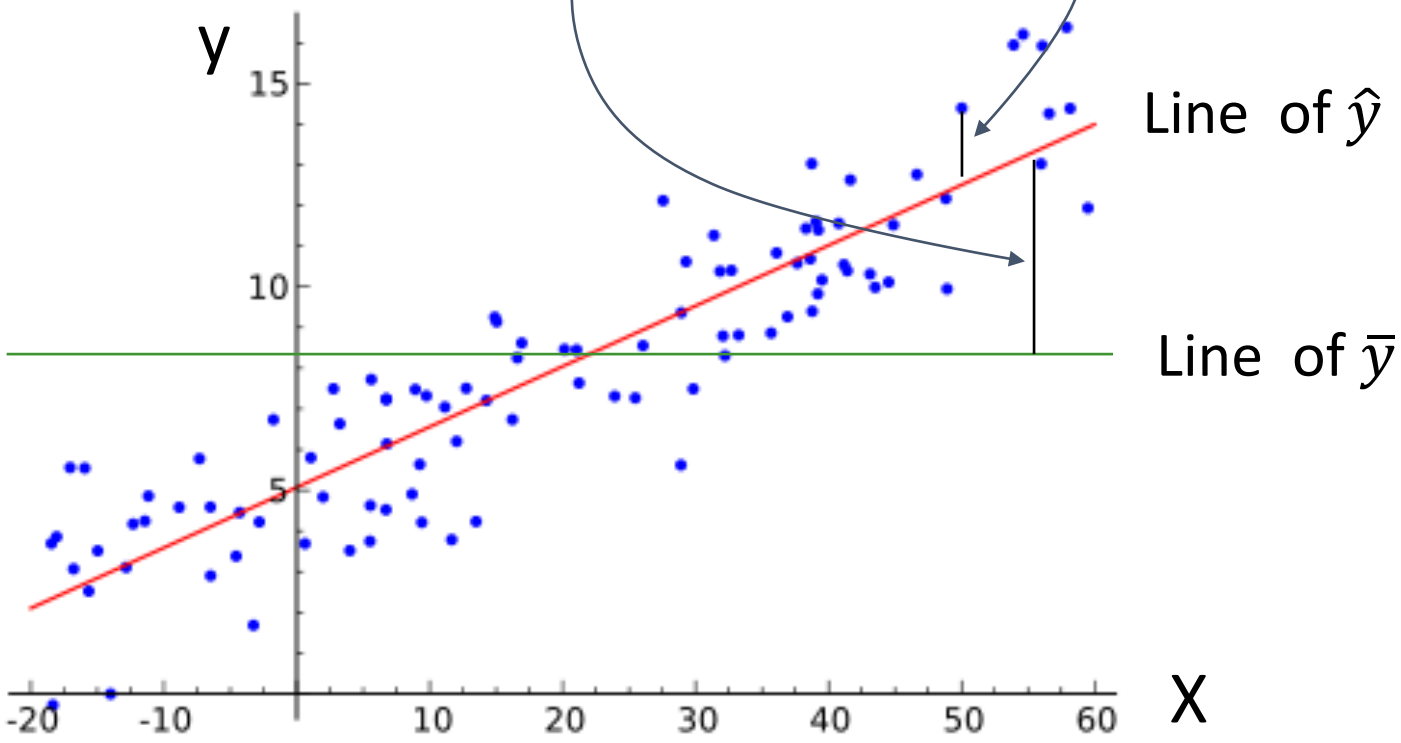
$$R^2 = 1 - \frac{\sum(y_i - \widehat{y_i})^2}{\sum(y_i - \bar{y})^2}$$

- The score is between 0 and 1
  - A value of 0 corresponds to a constant model that predicts the mean value of all training target values
  - A value of 1 corresponds to perfect prediction
- Also known as "coefficient of determination"

R-squared

Small if good fit

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$



R^2-values and p-values

- Use a hypothesis test
  - The null hypothesis: there is no linear relationship ($\beta = 0$)
  - Some value which should be small under the null hypothesis, and large if the alternate hypothesis is true
- Use R-squared value and F-distribution to get p-value
  - From R-squared we can derive another statistic (using degrees of freedom) that has a standard distribution called an F-distribution
  - From the CDF for the F-distribution, we can derive a p-value for the data
- If p is small, say less than 0.05, we conclude that there is a linear relationship

ANY QUESTIONS?